

Observational Indistinguishability and Integrity Blind Regions Across the Evidence Boundaries of Hybrid Quantum-Classical Kernel Workflows

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Abstract—Integrity auditing asks whether the evidence exposed at a workflow boundary can distinguish a consequential intervention from the baseline. We formalize this question for hybrid quantum-classical workflows as observational indistinguishability under an explicit information set and explicit trusted references. A structural blind region is the set of interventions whose observable view equals the baseline view; it is monotone under refinement of the view and is closed exactly by added evidence that separates the intervention class. The evaluation-label path is a protected boundary: for a fixed predictor, label-only changes leave feature and prediction evidence invariant, prior-preserving changes also leave label marginals invariant, and any change of the reported balanced accuracy changes the confusion matrix, so a material label corruption is always separable in the joint item-label view; a trusted aggregate reference of the same batch detects it exactly, and item alignment is needed only for aggregate-preserving relabelings. Across CICIDS2017, UNSW-NB15 and ToN-IoT in eight fixed environments (11,400 observations, 3,600 label interventions) the blind regions hold without exception and survive null calibration. A conformal family rule with an exact finite-sample level calibrates the regimes with feature evidence to 0.053–0.058 against 0.125–0.259 for the union of per-sensor rules. An offline end-to-end `allow/hold/block` evaluation over 24,000 frozen observations shows that batch-level regimes with feature evidence serve 4,365–4,496 of 7,008 materially changed results; the trusted-reference regime serves none but interrupts 0.52 of benign near-null variation. An adaptive attacker who preserves the tight feature clusters that the batch-level sensors rely on cuts their detection of feature drift from 0.96–1.00 to 0.01–0.66 while keeping 83–91% of the conclusion changes. A 165-cell simulator gate maps the same lattice onto circuit, kernel and estimation evidence, and a four-contract prototype executes the composed decision.

Index Terms—hybrid quantum-classical systems, integrity auditing, observability, quantum kernels, runtime assurance

1 INTRODUCTION

HYBRID quantum-classical software distributes one reported result across heterogeneous computational and evidential boundaries. Classical records are acquired, cleaned, projected and encoded; a circuit and an execution stack

produce quantum evidence; a classical learner turns a kernel into a decision; and an evaluation layer aligns that decision with reference outcomes and reports a metric. A plausible final number may therefore coexist with a failure in the data, circuit, kernel, prediction, label or reporting path, and security analyses of hybrid systems call for lifecycle-aware controls rather than trust in an isolated algorithm [1], [2].

Robustness benchmarking collapses this chain into a performance change. That endpoint cannot say what changed or whether the available evidence could have revealed the change. Consider the evaluation boundary: a deployed predictor receives the same feature batch and emits the same predictions, but the labels used to score the batch are corrupted. Balanced accuracy moves although the model has not. Feature and prediction monitors cannot observe the event because their inputs are identical; label counts expose an unbalanced corruption, but a balanced exchange of class identities preserves them. Calling the event stealthy omits the decisive qualifier: stealthy to which auditor, holding which evidence, trusting which reference?

That qualifier has a long history outside machine learning. Control theory characterizes undetectable and stealthy attacks as inputs that are indistinguishable from nominal behaviour through the monitor’s outputs [3]–[5], quantifies the detectability–impact trade-off [6], [7], and treats observability as a property of the monitored system rather than of the attack [8]. This article transports that discipline to the evidence chain of a hybrid learning workflow and makes three moves that the control-theoretic literature does not: it treats the *evaluation-label path* as a protected boundary with its own blind regions; it separates structural indistinguishability (the view is identical) from statistical undetectability (the view differs but not beyond the null variability of fresh batches) and shows which *trusted reference* of the same batch converts the second into an exact test (an aggregate for conclusion integrity, an item-aligned one for item identity); and it composes the resulting coverage into an executable, calibrated decision whose costs are measured on both sides.

The contribution is one package:

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- an observation model of hybrid workflows in which auditability is a property of (intervention class, information set, trusted references), with monotonicity, closure and materiality results whose earlier label-path

propositions are corollaries, minimal counterexamples that separate integrity violation, marginal invariance, confusion-matrix invariance and conclusion change, and the quantum branch as an instance of the same view lattice;

- an explicit adversary and failure model that assigns every executed intervention to fault robustness, integrity corruption, adversarial attack or adaptive attacker, with the roots it cannot write, including an executed adaptive attacker who preserves the fingerprint that the batch-level sensors rely on;
- multi-dataset, fixed-OOD validation with verified deduplication, cluster-level uncertainty and preregistered null calibration in which the exact blind regions survive;
- a decision-level calibration by a conformal family rule with an exact finite-sample level under exchangeability, and an offline end-to-end `allow/hold/block` evaluation on 24,000 frozen observations whose primary endpoint is the number of materially changed results a policy would serve and whose secondary endpoint is what each policy costs on benign variation; and
- a bounded quantum-specific layer and a four-contract prototype that execute the composed decision within a local research boundary.

No constituent is claimed to be individually sufficient novelty. The propositions are finite-batch statements; the family rule is a standard conformal construction; contracts, hash chains and semantics-preserving tests have clear prior art; and the secondary quantum-versus-classical impact profile, reported in the supplement, is neither a quantum-advantage result nor a vulnerability ordering. Physical hardware, calibrated backend noise, scheduling, provider-side security, multi-tenant interference, deployed runtime services and operational Fleet Management are outside the claim.

2 RELATED WORK AND POSITIONING

2.1 Stealth, Detectability and Observability in Secure Systems

The idea that detectability depends on what the monitor can see is established. Pasqualetti *et al.* characterize undetectable and unidentifiable attacks on cyber-physical systems through the observability of the monitored dynamics [3]; Teixeira *et al.* organize adversaries by model knowledge, disclosure and disruption resources and derive zero-dynamics and replay attacks that are stealthy to residual detectors [4]; Liu *et al.* exhibit false-data injections that bypass bad-data detection because they lie in the estimator’s range space [5]. Later work quantifies how much an ϵ -stealthy attacker can degrade performance and how to limit that impact [6], [7], and surveys physics-based detection as a question of which residuals a monitor can form [8]. Runtime assurance routes a decision through a verified simple controller when a monitor flags the complex one [9], runtime verification checks executions against specifications [10], and integrity checkers compare stored digests against trusted baselines [11].

This article does not claim the general insight that detectability is information-relative. It differs in the object and in the missing pieces: the monitored system is an evidence chain whose protected boundaries include the ground truth used to score the model; the blind regions are derived from views rather than from linear dynamics; the role of a trusted reference is made explicit as the difference between statistical and exact separation; and the coverage is composed into a calibrated decision and evaluated end to end on frozen outputs, including against an adaptive attacker.

2.2 Monitoring, Labels and Evaluation Integrity

Label shift estimation targets a changed class prior under source-target assumptions [12]; MLDemon and uncertainty-aware monitoring combine unlabeled observations with selectively acquired labels [13], [14]; an evidence-sufficiency framework maps proxy coverage under delayed ground truth and formalizes the inability of feature proxies to expose concept drift [15]. Northcutt *et al.* document pervasive label errors in test sets and their effect on benchmark rankings [16]; Arp *et al.* catalogue sampling, labeling, metric and threat-model pitfalls in security benchmarks [17]; Bajaj names *evaluation blindness*, a measurement indistinguishable from a healthy state while the system fails, and documents its prevalence in production incidents [18]. These works treat labels as available, noisy or late. Here the label path is an asset under intervention, availability is separated from trust, and the evidence needed to identify a corruption is derived, not assumed.

2.3 Calibration of Multi-Sensor Decisions

The family rule of Section III-D is a conformal p -value [19], [20] built on the maximum of per-sensor rank scores, i.e. on Tippett’s minimum- p combination [21] with the maximum-statistic calibration of Westfall and Young [22]. Its finite-sample level under exchangeability follows the standard exchangeability argument, which holds with ties when ties are counted against rejection [19], [23]; conformal anomaly detection applies the same construction to novelty scores [24], and Bates *et al.* analyse the dependence among conformal p -values that share one calibration set and the resulting multiple-testing behaviour [25]. We claim no novelty for the rule. Its role is to give the decision of an information regime a false-alarm level that is exact under a stated premise, and to make the violation of that premise by the executed design measurable. The asymmetric variant of the rule that an earlier version of this evidence used had no such guarantee (Section III-D, Proposition 5(c)).

2.4 Hybrid and Quantum Software Integrity

Volya *et al.* decompose classical-quantum systems into control, compilation, execution and measurement surfaces [1]; surveys extend the threat space to malicious compilation, hardware faults, cloud exposure and side channels [2], [26]; Quantum Vulnerability Factor scores circuit-output sensitivity to faults [27] and Quantum Leak demonstrates timing leakage in cloud services [28]. QCIVET combines stage contracts, hash-chained traces, behavioural subtyping and calibrated observable-deviation tests with QPU

TABLE 1

Positioning against the closest lines of work. A dash means that the property is not the work’s declared object, not that the work is deficient.

Line of work	Evidence conditioning	Evaluation-label boundary	Trusted reference	Validation and response
Stealthy attacks on CPS [3]–[6]	Observability of linear dynamics	–	Model and sensors trusted	Residual detectors; impact bounds
Runtime assurance and integrity monitoring [9]–[11]	Specification or baseline digest	–	Verified controller or stored baseline	Switching, monitors, digests
Hybrid threat taxonomies [1], [2]	–	–	–	Taxonomy; no executable response
QCIVET [29]	Stage contracts/observables	–	Reference observables	Calibrated noise and QPU; verification engine
QML-PipeGuard [30]	Calibrated observable tolerance	–	Fingerprint of the channel	Two-qubit QSVM on IBM Heron
Evidence sufficiency [15]; evaluation blindness [18]	Proxy coverage; detectability taxonomy	Delayed or silent outcomes	–	One dataset; incident corpus
This work	Views, refinement order, structural versus calibrated blindness	First-class protected boundary	Explicit; exact versus statistical separation	Three datasets, eight environments, conformal policy evaluated offline end to end, adaptive attacker

validation [29]; QML-PipeGuard fingerprints the quantum stage through expectation values under a tomographically structured measurement family, absorbs benign calibration drift within a calibrated tolerance and detects channel substitution on an IBM Heron processor [30]; a multi-level framework evaluates structural, interaction and behavioural circuit-integrity metrics [31]; and a cross-provider analysis proposes a unified provenance contract for quantum jobs [32]. Quantum software testing addresses the oracle problem through differential, metamorphic and equivalence-modulo-input relations [33]–[35], and compiler verification, program logics and runtime assertions supply formal and runtime mechanisms [36]–[40]. We make no priority claim for contracts, hash chains, stage decomposition, semantic probes or fail-closed verification. Their unit is the quantum stage or the pipeline trace; ours is the information set that makes a changed asset distinguishable, with the evaluation-label path as a first-class boundary, coverage mapped across data environments, and the contract executing a calibrated map.

Table 1 states the combination claim. Prior work establishes information-relative detectability for controlled dynamics, proxy monitoring for late labels, and contract-based integrity for quantum stages. This work differs in deriving blind regions from views of a learning workflow whose protected boundaries include the evaluation labels, in separating structural from statistical blindness through explicit trusted references, and in composing the coverage into a decision whose false-alarm level, unsafe-allow count and benign-interruption cost are measured.

2.5 Relation to Companion Work

Three companion studies by the authors share infrastructure or vocabulary and are disclosed to delimit the claim. *Sharp target-domain certificates* [41] bounds the advantage of a fixed quantum-kernel candidate under distribution shift with partial target labels; it shares data sources, fidelity kernels and a fail-closed prediction lock, but its object is an identification interval, not the observability of interventions. *Conditional validity of quantum event classifiers* [42] is the closest conceptual relative: it conditions fail-closed verdicts on a declared information set and exhibits exact sensor

blindness to weight-only nuisances in collider data. The present article applies the conditioning to the integrity of the evidence chain under controlled interventions, with trusted references, calibrated policy and executable response. *Candidate comparability before promotion* [43] decides whether a challenger should replace a deployed detector after a drift alarm; it shares the benchmarks and label-free monitors, but its unit is a sequential promotion decision. None of the propositions, interventions, tables or experiments reported here appears in those works.

3 OBSERVATION MODEL AND BLIND REGIONS

The complete statement with proofs is in the supplement; this section states the objects and the results the evidence tests. The results are elementary by design: their role is to make exact which evidence separates which intervention class, so that the gates test statements rather than intuitions.

3.1 States, Interventions and Materiality

A workflow state is the tuple of artifacts of one evaluation run,

$$s = (X, P, C, K, E, f, y, R) \in \mathcal{S}, \quad (1)$$

with acquired features X , preprocessing P , circuit or feature map C , estimated kernel K , execution metadata E , fitted predictor f , evaluation labels y and reported result R ; $\tilde{X} = P(X)$ is the evaluation representation and $\hat{y} = f(\tilde{X})$ the predictions, indexed by item identity $i = 1, \dots, n$. An intervention is a map $a : \mathcal{S} \rightarrow \mathcal{S}$; the baseline is s_0 and the intervened state $s_a = a(s_0)$. Classes used below: label-only T_y (changes y item-wise, fixes everything else), its prior-preserving subclass T_y^π (also preserves the class histogram), and feature-side mechanisms that change $\tilde{X}$ and, through f , possibly $\hat{y}$ and R . The conclusion is $R(s)$, balanced accuracy of $\hat{y}$ against y , and its signed change is

$$\Delta_R(a) = R(s_0) - R(s_a). \quad (2)$$

An intervention is *material at level τ* if $|\Delta_R(a)| \geq \tau$; $\tau \rightarrow 0^+$ means any change of the reported conclusion. Earlier versions of this evidence used the positive part $\max(\Delta_R, 0)$;

the signed endpoint is used here because an apparent improvement caused by an integrity failure is a conclusion change.

3.2 Views, Refinement and Trusted References

An information set $\mathcal{I}$ is a view map $V_{\mathcal{I}} : \mathcal{S} \rightarrow \mathcal{O}_{\mathcal{I}}$; the auditor observes $V_{\mathcal{I}}(s)$ and nothing else. The regimes studied are $\mathcal{I}_X$ (the multiset of rows of $\tilde{X}$), $\mathcal{I}_{XF}$ (the multiset of pairs $(\tilde{x}_i, \hat{y}_i)$), $\mathcal{I}_{Y_m}$ (the class-count vector of y), $\mathcal{I}_{XFY}$ (the multiset of triples $(\tilde{x}_i, \hat{y}_i, y_i)$) and $\mathcal{I}_Q$ (circuit, kernel and execution evidence). $\mathcal{I} \subseteq \mathcal{I}'$ (refinement) iff $V_{\mathcal{I}} = \pi \circ V_{\mathcal{I}'}$ for some π ; hence $\mathcal{I}_X \subseteq \mathcal{I}_{XF} \subseteq \mathcal{I}_{XFY}$ and $\mathcal{I}_{Y_m} \subseteq \mathcal{I}_{XFY}$, while $\mathcal{I}_{Y_m}$ is incomparable with $\mathcal{I}_X$ and $\mathcal{I}_Q$ is a separate branch. The join $\mathcal{I} \vee \mathcal{W}$ is the pair of views.

A *reference* is a stored value $\rho = V_{\mathcal{J}}(s_0)$ of a view of the baseline. It is *trusted* under the assumption that no intervention in the class under study can alter it. Three levels of evidence are distinguished throughout: (A) batch-level statistical evidence, with no stored baseline value; (B) a trusted *aggregate* reference, a permutation-invariant summary of the same batch such as the confusion matrix $M(s_0)$, $\text{hist}(y_0)$ or $R(s_0)$; (C) a trusted *item-aligned* reference, an item-indexed tuple such as $(y_{0,i})_i$. Correspondingly, for a label-path intervention, *item-identity integrity* holds if $y_a = y_0$ item-wise, *aggregate integrity* if $M(s_a) = M(s_0)$ and *conclusion integrity* if $R(s_a) = R(s_0)$; each implies the next and no converse holds. $\mathcal{I}^*$ denotes a regime augmented with trusted references of its own baseline view ($\mathcal{I}_{XFY}^*$ holds both the aggregate confusion profile and the item-aligned labels of the same batch). A label being available in $\mathcal{I}_{XFY}$ says nothing about whether it is trusted: if the label store is the asset under attack, the auditor sees y_a , not y_0 .

3.3 Equivalence, Sensors and Blind Regions

Definition 1. $s \sim_{\mathcal{I}} s'$ iff $V_{\mathcal{I}}(s) = V_{\mathcal{I}}(s')$. A sensor admissible in $\mathcal{I}$ is a measurable $S : \mathcal{O}_{\mathcal{I}} \rightarrow \mathbb{R}^k$, written $S(s) = S(V_{\mathcal{I}}(s))$.

Lemma 1 (structural blindness). If $s_a \sim_{\mathcal{I}} s_0$ then $S(s_a) = S(s_0)$ for every sensor admissible in $\mathcal{I}$ (path-wise for a seeded randomized sensor; in distribution for a fresh seed).

Definition 2. For a class $\mathcal{A}$, the structural blind region of $\mathcal{I}$ is $B_{\mathcal{I}}(\mathcal{A}) = \{a \in \mathcal{A} : a(s_0) \sim_{\mathcal{I}} s_0\}$ and the sensor blind region of a family $\mathcal{S}$ is $B_{\mathcal{I},\mathcal{S}}(\mathcal{A}) = \{a : S(a(s_0)) = S(s_0) \forall S \in \mathcal{S}\}$. By Lemma 1, $B_{\mathcal{I}} \subseteq B_{\mathcal{I},\mathcal{S}}$, with equality iff $\mathcal{S}$ is *baseline-separating* on the orbit of $\mathcal{A}$, i.e. takes a different value at $a(s_0)$ than at s_0 whenever the views differ. Baseline separation is what detection needs; identifying *which* intervention occurred would need pairwise separation of intervened states, which is not claimed. Sensor blindness thus decomposes into information-level blindness and sensor insufficiency; the adaptive attacker of Section VI-D exploits the second. The structural and sensor auditability gaps at level τ are

$$G_{\mathcal{I}}(\tau) = \{a : |\Delta_R(a)| \geq \tau, a \in B_{\mathcal{I}}\}, \quad G_{\mathcal{I},\mathcal{S}}(\tau) \supseteq G_{\mathcal{I}}(\tau). \quad (3)$$

Proposition 1 (monotonicity). If $\mathcal{I} \subseteq \mathcal{I}'$ then $B_{\mathcal{I}'}(\mathcal{A}) \subseteq B_{\mathcal{I}}(\mathcal{A})$ and $G_{\mathcal{I}'}(\tau) \subseteq G_{\mathcal{I}}(\tau)$ for every τ .

Proof: $V_{\mathcal{I}'}(s_a) = V_{\mathcal{I}'}(s_0)$ implies $V_{\mathcal{I}}(s_a) = \pi(V_{\mathcal{I}'}(s_a)) = \pi(V_{\mathcal{I}'}(s_0)) = V_{\mathcal{I}}(s_0)$. $\square$

Corollary 1 (label-path boundaries). Let f be fixed and deterministic. (a) $T_y \subseteq B_{\mathcal{I}_{XF}} \subseteq B_{\mathcal{I}_X}$: every sensor admissible in $\mathcal{I}_X$ or $\mathcal{I}_{XF}$ is invariant under a label-only change. (b) $T_y^\pi \subseteq B_{\mathcal{I}_{Y_m}}$: every sensor admissible in $\mathcal{I}_{Y_m}$ is invariant under a prior-preserving change. These are the two propositions of the earlier version of this evidence, now instances of Lemma 1 and Proposition 1.

Proposition 2 (closure). $B_{\mathcal{I} \vee \mathcal{W}}(\mathcal{A}) = B_{\mathcal{I}}(\mathcal{A}) \cap B_{\mathcal{W}}(\mathcal{A})$. A class $\mathcal{A} \subseteq B_{\mathcal{I}}$ becomes completely separable after adding $\mathcal{W}$ iff $V_{\mathcal{W}}(a(s_0)) \neq V_{\mathcal{W}}(s_0)$ for every $a \in \mathcal{A}$: the added evidence must be baseline-separating on the orbit of the class. Consequently no view factoring through $(\tilde{X}, f(\tilde{X}), \text{hist}(y))$ separates any $a \in T_y^\pi$; the multiset $\mathcal{I}_{XFY}$ separates it unless labels are permuted only among items with identical $(\tilde{x}, \hat{y})$, and $\mathcal{I}_{XFY}^*$ separates every non-identity relabeling.

Proposition 3 (materiality forces aggregate separability). Let $R = g(M)$ be a function of the confusion matrix $M(s)$, as balanced accuracy is. If $a \in T_y$ and $\Delta_R(a) \neq 0$ then $M(s_a) \neq M(s_0)$; hence a is separable in the confusion view and in $\mathcal{I}_{XFY}$, and $G_{\mathcal{I}_{XFY}}(\tau) \cap T_y = \emptyset$ for every $\tau > 0$.

Proof: Contrapositive of $M(s_a) = M(s_0) \Rightarrow R(s_a) = R(s_0)$; the confusion view is a coarsening of the multiset of triples. $\square$

The proposition uses only $R = g(M)$ and holds for every metric that is a function of the binarised confusion matrix. A ranking metric such as ROC-AUC is not: it depends on the multiset of (score, label) pairs, and the same argument then holds with that multiset as the aggregate (supplement). No AUC experiment is run.

3.4 Three Auditor Classes and Decision-Level Calibration

A *reference-anchored* auditor holds a trusted reference ρ (level B or C) and computes $D(s) = d(V_{\mathcal{J}}(s), \rho)$ with a metric d ; then $D(s_0) = 0$ exactly, the rule “fire iff $D > 0$ ” has zero false-alarm probability, and separability in $\mathcal{J}$ is sufficient for detection. An *anchor-free* (batch-level) auditor holds only a null model of $V_{\mathcal{I}}$ over fresh clean batches and decides with a calibrated test at budget α ; for a separable a its detection probability is the power against the shift relative to batch-to-batch variability, and separability is not sufficient. A *post-hoc certifier* recomputes a view from inputs it trusts (the evaluation contract recomputes R from $(\hat{y}, y)$) and needs trusted inputs rather than a stored value.

Proposition 4. (i) For any auditor whose decision is a function of $V_{\mathcal{I}}(s)$, $a \in B_{\mathcal{I}}$ implies $\text{fire}(s_a) = \text{fire}(s_0)$ on every batch, so the calibrated detection rate equals the false-alarm rate. (ii) A reference-anchored auditor detects every $a \notin B_{\mathcal{J}}$, deterministically.

Corollary 3 (which reference certifies which integrity). Let $a \in T_y$ with f fixed. (a) With a trusted aggregate reference $M_0 = M(s_0)$ of the same batch, “fire iff $M(s_a) \neq M_0$ ” detects every violation of aggregate integrity and, by Proposition 3, every material a : conclusion and aggregate integrity need only level B. (b) Every reference that factors through M , $\text{hist}(y)$ or R is invariant under relabelings that permute labels among items with equal predictions (C1), which violate item-identity integrity; certifying item identity needs level C, and an item-aligned y_0 detects every non-identity relabeling.

Proposition 5 (union versus conformal family calibration). (a) If m sensors fire under the null with probabilities

$p_j \leq \alpha$, then $\max_j p_j \leq \Pr_0[\bigcup_j E_j] \leq \min(1, \sum_j p_j) \leq \min(1, m\alpha)$, with the bounds attained by nested and disjoint events and $1 - (1 - \alpha)^m$ under independence. (b) Let $v_1, \dots, v_n$ be the sensor vectors of n calibration draws and v_{n+1} that of the audited batch; give every member j of the augmented set the leave-one-out family score $\tilde{U}_j = \max_s \#\{i \neq j : v_{s,i} < v_{s,j}\}/n$ against the other n members, let $p = (1 + \#\{i \leq n : \tilde{U}_i \geq \tilde{U}_{n+1}\})/(n+1)$, and fire iff $p \leq \alpha$ (ties against firing). For every augmented set at most $\lfloor \alpha(n+1) \rfloor$ members would fire if audited; hence, if the $n+1$ draws are exchangeable, $\Pr_0[p \leq \alpha] \leq \lfloor \alpha(n+1) \rfloor / (n+1) \leq \alpha$ with no continuity or tie assumption [19], [21], [22]; $10/201 = 0.0498$ for $n = 200$, $\alpha = 0.05$. For one sensor without ties the rule is the per-sensor rule of Gate N. The level is marginal over the joint draw of calibration set and audited batch; nothing is claimed when the premise fails. (c) The asymmetric rule of the earlier version, which scored calibration draws against the other $n-1$ calibration draws only, has no such guarantee: with three sensors and cyclic orderings over five vectors it fires with probability 0.60 at $\alpha = 0.2$ under exchangeability (supplement), and on the frozen draws it exceeds α under exchangeable re-splits where the conformal rule does not (Section VI-B).

Proposition 6 (no local authentication without an uncontrolled root). Let a local verifier be any map $\text{acc}(V_{\mathcal{I}}(s), \rho)$ of the observed view and a reference bundle, with honest references $\rho_H(s) = V_{\mathcal{J}}(s)$ on an honest family $\mathcal{S}_H$. If the verifier accepts every honest pair (honest completeness) and the class can produce $(V_{\mathcal{I}}(s'), \rho_H(s'))$ for some honest $s' \not\sim_{\mathcal{I}} s_0$ (joint control), the substitution is accepted: it is observationally an honest run of s' , so no function of that input establishes authenticity relative to s_0 , and if $R(s') \neq R(s_0)$ a material change is served. Conversely, a bundle component $V_{\mathcal{K}}(s_0)$ the class cannot write rejects every substitution with $V_{\mathcal{K}}(s') \neq V_{\mathcal{K}}(s_0)$ (Proposition 4(ii)). Authenticity relative to a baseline thus requires at least one root outside the declared class, which suffices exactly for the classes it separates; the result is about authenticity, not consistency, since by Proposition 3 a material substitution does change the joint view.

3.5 Counterexamples

Table 2 lists minimal constructions that separate the events a robustness number conflates, with the number of frozen expansion observations that realise each of them. C1 is an integrity violation with no conclusion impact, visible only item-wise against a trusted reference; C2–C4 separate marginal invariance, confusion-matrix invariance and conclusion invariance; C5–C6 do the same on the feature side; C7 shows that the clipped “harm” endpoint of the earlier evidence hid 1,534 conclusion changes, 433 of them on the label path; C8 is the empirical face of Proposition 3. Of the 3,600 label rows, a trusted aggregate reference of the same batch detects 2,792 (every material one, Corollary 3a), and the remaining 808 are C1 relabelings that only an item-aligned reference exposes (Corollary 3b). A ninth construction, the cluster-preserving drift of Section VI-D, is not a structural blind region but sensor insufficiency against an adaptive attacker, and is measured rather than proved.

TABLE 2
Minimal counterexamples and their witnesses in the frozen expansion (count / denominator). $\hat{y}$ predictions, y labels, M confusion matrix, R balanced accuracy.

ID	Construction and what it separates	Witnesses
C1	Swap y_i, y_j with $\hat{y}_i = \hat{y}_j$: M , hist, R invariant; item identity changes	808 / 3,600
C2	Relabeling that changes M : $\hat{y}$ invariant, R changes	2,617 / 3,600
C3	Swap y_i, y_j with $\hat{y}_i \neq \hat{y}_j$: hist invariant, M and R change	1,059 / 1,800
C4	Compensating flips: M changes, R invariant	175 / 3,600
C5	Feature change that flips predictions but not R	296 / 7,200
C6	Feature change that crosses no decision boundary	2,513 / 7,200
C7	Intervention that raises R (apparent improvement)	1,534 / 10,800
C8	Material label change always changes M (Prop. 3)	2,617 / 2,617

3.6 The Quantum Branch as an Instance of the Lattice

Let $\mathcal{C}$ be the circuit representations (canonical OpenQASM 3 with bound parameters), $\Phi : \mathcal{C} \rightarrow \mathcal{K}$ the semantic map to the ideal fidelity kernel on a fixed probe set, $h(C)$ the provenance hash, $A(K)$ the algebraic invariants, $f_K(\tilde{X})$ the decision computed from a kernel and $\hat{K} \sim P(\cdot | C, E)$ the finite-shot estimator under execution context E .

Proposition 7 (quantum lattice). Assume h collision-free on the circuits under study. (i) $[C] = \Phi^{-1}(\Phi(C))$ is the semantic class of C ; approved transpilation and common-unitary rewrites map C into $[C]$ without fixing $h(C)$, so provenance refines the kernel view, $B_h \subseteq B_K$, strictly whenever the class contains an approved rewrite: separability in provenance is not harm, and the auditor needs the approved class, implemented as semantic equality of probe kernels against the trusted $K_0 = \Phi(C_0)$ (level B). (ii) A and f_K coarsen K , so $B_K \subseteq B_A$ and $B_K \subseteq B_f$; a PSD-preserving substitution with $A(K') = A(K_0)$ lies in $B_A \setminus B_K$ and is closed only by the anchored comparison with K_0 ; a kernel change that crosses no decision boundary lies in $B_f \setminus B_K$. (iii) Under shot noise $\Pr[\hat{K} = K_0] = 0$ even for the honest circuit, so “fire iff $\hat{K} \neq K_0$ ” has false-alarm probability one; $d(\hat{K}, K_0)$ must be treated as a level-A statistic with a null from repeated honest estimation, to which Proposition 5(b) applies with one sensor, and the auditor must abstain where no such null is calibrated. The proposition adds no quantum mechanics: it places the quantum sensors on the same lattice and names the two features the classical branches lack, an approved equivalence class coarser than provenance and an estimation noise that turns an exact anchor into a statistical one. Hash chaining of the audit envelope is a reference-anchored comparison over the record itself; it is tamper-evident but its root is not assumed uncontrolled, so Proposition 6 applies.

4 ADVERSARY AND FAILURE MODEL

Table 3 assigns every executed intervention to one of four classes and states the roots it cannot write and the regime that identifies it; the complete per-class record (actor, capability, access point, knowledge, objective, budget, alterable assets, observable set, supported claim) is in the artifact. Evaluation-label classes stand for corruption of the ground-truth store or of the label join; feature-side classes for corruption or drift of the acquisition path feeding both branches; the cluster-preserving class for an attacker who knows which batch-level fingerprint the auditor relies on; circuit and kernel

TABLE 3

Adversary and failure model of the executed interventions. Roots: assets the class cannot write. Identifying regime: the least evidence that separates the class in the frozen design under the conformal family rule; “exact” means an item-aligned trusted reference.

Mechanism (boundary)	Class	Capability	Roots it cannot write	Identifying regime	Supported claim
Random label flip (evaluation/report)	Integrity corruption	cor-rewrite r of y at random	X, P, C, K, f ; item-aligned y^* if declared	exact only; $\mathcal{I}_{Y_m}/\mathcal{I}_{X_{FY}}$ batch 0.00–0.01	invisible to $\mathcal{I}_X, \mathcal{I}_{X_F}$ (Cor. 1a); not detected at budget by marginal or batch joint evidence at these rates
Prior-preserving label flip (evaluation/report)	Adaptive attacker	pairwise swaps preserving class counts	as above	exact only; $\mathcal{I}_{X_{FY}}$ batch 0.00–0.01	invisible to $\mathcal{I}_X, \mathcal{I}_{X_F}, \mathcal{I}_{Y_m}$ (Cor. 1); every material instance changes M (Prop. 3)
Confusion-preserving relabeling (realised, not executed as a suite)	Adaptive attacker	swaps among equal predictions	as above	item-aligned reference only	item-identity violation without aggregate or conclusion impact (C1; 808 rows)
Feature sign flip (acquisition/preprocessing)	Integrity corruption	cor-negate p of $\tilde{X}$ entries	training data, fitted model, reference features	$\mathcal{I}_X$ 0.09–0.72; $\mathcal{I}_{X_F}$ 0.13–0.84; exact 0.79–0.91	partially separable; residual is strength-dependent
Mean shift, scaling drift (acquisition/preprocessing)	Fault robustness	robust-shift or scale all features	as above	$\mathcal{I}_X$ 0.96–1.00 by cluster fingerprint	detected in this design by a fragile fingerprint
Feature dropout with imputation (preprocessing)	Fault robustness	robust-impute p of entries	as above	batch 0.00–0.03; exact 0.49–0.76	separable in $\mathcal{I}_{X_F}$ yet statistically undetectable without a reference
Cluster-preserving mean shift and scaling (executed, Gate A)	Adaptive attacker	same mechanisms on the 32% of entries outside tight clusters	as above	$\mathcal{I}_X$ 0.01–0.56, $\mathcal{I}_{X_F}$ 0.06–0.66 at matched strengths; exact for every material row	evades the fingerprint while keeping 88–95% of the conclusion changes; served by P2 in 28–39% of material rows
Circuit-parameter mutation; repetition change (design/transpilation)	Adversarial attack	alter C before execution	reference circuit hash, probe kernel K_0	provenance and semantic 15/15	identified by reference-anchored provenance and semantics; outputs miss sub-decision changes
Benign transpilation; common-unitary rewrite	Approved equivalence	compiler pass	–	hash changes 15/15, K unchanged	separability in $h(C)$ is not harm (Prop. 7)
Asymmetric, diagonal-eroded kernel edit (post-processing)	Integrity corruption	cor-edit K after estimation	reference kernel, algebraic invariants	algebra 15/15	detected by structural invariants without a reference
PSD-preserving kernel substitution (post-processing)	Adaptive attacker	valid-looking K	reference kernel, circuit hash	algebra 0/15; provenance and semantic 15/15	sensor-level blindness closed by reference anchoring (Prop. 7)
Binomial shot emulation (execution)	Fault robustness	robust-finite-shot estimate	–	repeated estimation 15/15	held for adjudication; not hardware evidence
Post-hoc envelope edit (record)	Adversarial attack	rewrite stored contracts	none assumed	hash chain	tamper-evident only; no identity or non-repudiation

classes for corruption of the circuit/parameter store, the compiler output or the post-estimation kernel; shot emulation for legitimate estimation uncertainty. Equal weighting is a stress profile, not a threat distribution, and no prevalence is estimated.

The evidence trusts the local runtime and host, the training data and fitted model, the reference input and preprocessing declaration, the reference circuit and probe kernels, SHA-256 collision resistance, the item-level label reference only where $\mathcal{I}_{X_{FY}}^*$ is declared, and the code that builds and verifies the envelope. Compromise of the operating system, runtime or verifier; provider identity, forged job or calibration metadata and real provider attestation; malicious schedulers, physical mapping and unapproved passes on hardware; device-calibrated noise, crosstalk, multi-tenant interference and QPU faults; side channels and circuit confidentiality; denial of service; and the operational Fleet Management System are outside this study.

5 METHODOLOGY

5.1 Data, Pipelines and Interventions

The frozen first gate uses a balanced binary subset of CICIDS2017 [44]: 3,000 rows, 77 numeric columns and

128/128 capped training/evaluation samples. The expansion adds eight fixed environments E1–E8 (supplement): five from CICIDS2017 (a 256/256 scale gate and four temporal source-target pairs), two from UNSW-NB15 [45] (a balanced ID subset and a temporal pair) and one from ToN-IoT [46] (a balanced ID subset), all at 128/128 unless stated and projected to dimensions 8, 10 and 12. CICIDS2017 therefore carries six of the nine environments including Gate 1; the environments are not a probability sample of deployments. Train-defined preprocessing is applied without evaluation labels: TruncatedSVD per branch, training-fitted standardization for the RBF support-vector classifier (SVC) and frozen angular scaling to $[0, 2\pi]$ for the fidelity-kernel branches (ZZ, PauliXYZ and a Z/PauliXZ-equivalent profile, one repetition). Gate 1 uses the Qiskit Machine Learning reference evaluator [47]; the expansion uses a tagged exact-statevector evaluator validated against it at 10^{-10} on unit matrices and by an end-to-end replay (supplement). Neither is shot-noise or QPU evidence.

The intervention suite has six mechanisms at strengths 0.02, 0.05 and 0.10: feature sign flip, feature-wise mean shift, scaling drift, feature dropout with median imputation, random evaluation-label flip and prior-preserving evaluation-label flip. Feature-only sensors are feature Jensen–

Shannon divergence (JSD), maximum mean discrepancy (MMD) and the Kolmogorov–Smirnov (KS) rejection rate, each computed against the clean reference batch of the same items; prediction-aware evidence adds score JSD, prediction-rate shift and prediction JSD; label-marginal evidence uses prior shift and label JSD; joint-outcome evidence uses two confusion-profile statistics. Reference-anchored sensors (confusion-profile deltas against a trusted aggregate of the same batch; prediction disagreement and item-level label mismatch against item-aligned references) exist only with a trusted reference of the same items and have an exact-zero null.

5.2 Experimental Units and Uncertainty

Gate 1 crosses five split seeds with four nested model seeds; its 7,980 raw rows contain 3,420 exact repeated SVC rows, leaving 4,560 unique observations. The expansion crosses the same five split seeds with two model seeds; all 360 job pairs completed and 13,680 raw rows contain 2,280 verified repeats, leaving 11,400 unique observations. Coverage endpoints are exact counts in the prespecified design. For the secondary model-profile endpoint (supplement) the five split seeds are the inferential units with two-sided 95% Student- t intervals (four degrees of freedom), describing within-environment resampling only.

5.3 Null Calibration and Decision-Level Calibration

Gate N (preregistered in artifact 1.1.0) calibrates the ten distributional sensors per (environment, dimension, model) cell at a nominal per-sensor $\alpha = 0.05$: the threshold is the 191st of 200 clean calibration draws taken from one half of the held-out evaluation pool, and false alarms are measured on 200 draws from the disjoint other half. Gate F calibrates each regime as a family with the conformal rule of Proposition 5(b) at $\alpha = 0.05$ per decision, using the calibration draws only; the earlier asymmetric rule (Proposition 5(c)) and the union rule are computed on the same draws for comparison, and the pooled excess of the earlier rule over the nominal level is decomposed into rule bias (its rate minus the conformal rate on identical draws) and design effect (the conformal rate minus nominal), with both rules also evaluated under 30 exchangeable random re-splits of the 400 pooled draws of every cell. The 1.2.0 protocol, its amendment and the expectations recorded before regeneration are in the artifact. Because the 20 draws of a run are overlapping subsets of one pool half, the inferential unit for false-alarm rates is the (environment, split-seed) cluster: pooled rates over draws are descriptive, and intervals are five-cluster t intervals per environment. E1 (256-row draws from 322-row halves) is kept in the primary aggregate as preregistered and also reported separately.

5.4 Decision Layer and Offline End-to-End Evaluation

Gate D composes sensors, calibrated rules, the information regime, the trusted-reference status, a materiality threshold and the declared residual blind region into one `allow/hold/block` decision. `block` is reserved for violations of an exact invariant against a trusted reference; `hold` answers statistical evidence or, under the strict policy,

missing mandatory evidence; the decision composes with the four frozen contracts by the maximum in `allow < hold < block`. Four policies are fixed and named by what they are: P0 serve-always, the baseline; P1, the uncalibrated union rule of Gate N (risk-tolerant); P2, the conformal family rule, a calibrated *risk-tolerant* policy that may serve under an explicitly declared residual blind region; P3, the strict *fail-closed* (abstaining) variant that holds whenever a protected boundary (feature, prediction, label) is neither exactly nor statistically covered by the regime. The contracts of the prototype fail closed on their invariants; P2 does not, and is never called fail-closed. Five regimes are evaluated: $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$, $\mathcal{I}_{XFY}$ (batch-level) and $\mathcal{I}_{XFY}^*$ (trusted aggregate and item-aligned references of the same batch). The observation set is frozen: 12,000 clean evaluation draws, 1,200 near-null in-place controls (Gaussian noise $\sigma = 0.001$, scaling $\alpha = 0.001$), and the 10,800 intervened observations of the eight environments, 7,008 of which change the reported balanced accuracy (primary materiality; $\tau = 0.02$ and 0.05 as sensitivity). The primary endpoint is the unsafe `allow`: a material observation decided `allow`. The cost endpoints are the clean false-action rate (clean draws not allowed; for $\mathcal{I}_{XFY}^*$ the denominator is the 1,200 exact-zero rows) and the benign interruption rate (near-null controls held or blocked, the same 1,200 rows for every regime). The evaluation is offline, on frozen outputs; it is not a deployed runtime service. Twelve consistency checks fail closed, and every table and figure is generated from the manifested outputs.

5.5 Adversarial Gate: Cluster-Preserving Perturbation

Gate A (preregistered before execution) executes the adaptive attacker that the earlier version only discussed. The standardized projected features hold tight clusters of rows (in every environment some feature keeps 69–87% of the rows within 0.01 standard deviations) that a fresh clean batch reproduces and any in-place displacement smears; this is why mean shift and scaling drift were detected in every cell. The attacker holds the batch, knows the sensor definitions and the fingerprint, and applies the executed mechanisms unchanged to every entry that is not clustered, an entry being clustered when at least $\lceil 0.05n \rceil$ rows (never fewer than two) lie within 0.01 batch standard deviations of it; clustered entries and labels are untouched. Strengths 0.02, 0.05, 0.10 match the frozen suite and 0.25, 0.50 probe whether materiality can be bought on the unclustered entries; the executed mechanisms run in the same jobs with their frozen tags and seeds as matched controls and must reproduce the frozen expansion exactly. The gate runs in the eight environments with the frozen maps, caps and seeds (240 exact-statevector jobs) and is scored with the calibration draws, rules and policy layer of Gates F and D; window, minimum mass and strength grid were fixed before execution.

5.6 Quantum Gate and Executable Prototype

A separate simulator gate isolates the quantum boundary with CICIDS, 64/64 samples, five split seeds, dimensions 4, 6 and 8 and 11 conditions (165 cells): clean, benign level-1 transpilation and a common post-feature-map rewrite as controls; data-dependent RZ mutations at 0.02 and 0.10; a feature-map repetition change; asymmetric, diagonal-eroded

TABLE 4

Verified coverage counts after computational-unit deduplication. $\mathcal{I}_{Y_m}$ applies to prior-preserving rows only; the signed conclusion change is recomputed from the frozen tables by the artifact verifier.

Verified quantity	Gate 1	Expansion
Unique observations	4,560	11,400
Label-only / $\mathcal{I}_{XF}$ blind	1,440 / 1,440	3,600 / 3,600
Prior-preserving / $\mathcal{I}_{Y_m}$ blind	720 / 720	1,800 / 1,800
Lowered / unchanged / raised R	1,276/105/59	2,184/983/433
Changed R / joint response	1,335 / 1,335	2,617 / 2,617

and PSD-preserving kernel edits; and binomial fidelity-estimation emulators at 256 and 1,024 shots. Each cell records canonical OpenQASM 3 provenance, kernel hashes, semantic differences against a trusted reference, algebraic checks, repeated-estimation discrepancy and prediction changes. The prototype instantiates four ordered contracts (input/preprocessing, circuit/kernel, execution/result, evaluation/report) in an SHA-256 chain over one compact CICIDS cell and six prespecified scenarios; it evaluates composition and decision logic, not throughput.

6 RESULTS

6.1 Exact Label-Path Blind Regions

All 3,600 expansion and 1,440 Gate-1 evaluation-label observations have zero prediction disagreement and exactly unchanged feature JSD, MMD, KS rejection and score JSD; all 1,800 and 720 prior-preserving observations also have zero prior shift and label JSD (Table 4). The signed conclusion change is not one-sided: in the expansion 2,184 label observations lower the reported balanced accuracy, 983 leave it unchanged and 433 raise it (Gate 1: 1,276, 105, 59). Every one of the 2,617 label observations with a changed conclusion has non-zero item-aligned confusion evidence, and every one of the 2,184 with a lowered conclusion, the count reported by the earlier version, is among them. The boundary reproduces without exception across the eight environments and every model profile: the intervention has not become less stealthy; the auditor has acquired a more informative view; the supplement shows the same boundaries after per-sensor calibration.

6.2 Decision-Level Calibration

Over the 12,000 evaluation draws the per-sensor false-alarm rates lie between 0.029 and 0.069, but the batch-level rule “fire if any sensor fires” is not calibrated at the level of the decision: its rates are 0.125, 0.203, 0.048 and 0.259 for $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$ and $\mathcal{I}_{XFY}$, which join three, six, two (perfectly dependent) and ten sensors. The conformal family rule brings them to 0.056, 0.058, 0.048 and 0.053 (Fig. 1(a)); the per-environment cluster means range from 0.034 to 0.123 for $\mathcal{I}_{XFY}$. The asymmetric rule of the earlier version gave 0.061, 0.073, 0.048 and 0.079 on the same draws and attributed its whole excess over 0.05 to the design. That attribution was wrong: the rule bias is +0.005, +0.015, +0.000 and +0.025, the design effect +0.006, +0.008, -0.003 and +0.003, and under exchangeable re-splits of the pooled draws, where any excess is the rule’s own, the earlier rule fires at 0.052, 0.058, 0.040 and 0.059 while the conformal rule stays at 0.044, 0.042, 0.040 and 0.036 against

its level 0.0498. What remains with the conformal rule is the design effect, concentrated in E1, whose 200 draws per cell carry the information of about one independent batch per pool half (0.125, 0.115, 0.123 in $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{XFY}$); without E1 the conformal rule sits at 0.048, 0.052, 0.048 and 0.045. E1 stays in the primary aggregate. A split construction that ranks against one half of the calibration draws and calibrates on the other gives 0.049, 0.039, 0.067 and 0.034 (supplement) and is not adopted.

The exact blind regions are not threshold artifacts and not calibration artifacts: on all 3,600 label observations the per-sensor and the conformal $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ rules, and on all 1,800 prior-preserving observations the $\mathcal{I}_{Y_m}$ rules, fire zero times (Proposition 4(i)). The batch-level $\mathcal{I}_{XFY}$ rules detect 43 (union) and 11 (conformal) of the 2,617 material label observations, 0.016 and 0.004, although every one of them is separable (Proposition 3, C8): a 2–10% relabeling of a 128-row batch shifts the confusion profile by less than its variability across fresh clean batches. A trusted aggregate reference of the same batch detects all 2,617 exactly (Corollary 3a) and 2,792 of the 3,600 label rows in total; the 808 relabelings that preserve every aggregate (C1) are exposed only by the item-aligned label reference (Corollary 3b). Feature-side coverage is mechanism-dependent: mean shift and scaling drift are detected in 0.96–1.00 of cells because the standardized projected features hold tight clusters that an in-place perturbation smears and a fresh batch reproduces, a legitimate but fragile fingerprint that also fires on the near-null controls (40–64% with the conformal rule); feature dropout with imputation is nearly invisible at the batch level (0–3%) yet changes predictions in 49–76% of cells. Calibration costs little detection: containment of material observations is 0.40 under the union and 0.38 under the conformal rule in $\mathcal{I}_{XF}$ at the primary threshold, and 0.70 versus 0.66 at $\tau = 0.05$.

6.3 Offline End-to-End Decisions

Table 5 and Fig. 1(b,c) report the offline end-to-end evaluation. Under P0 every one of the 7,008 material observations is served. Under the batch-level regimes the conformal policy P2 serves 4,496 ($\mathcal{I}_X$), 4,365 ($\mathcal{I}_{XF}$) and 4,390 ($\mathcal{I}_{XFY}$) of them with clean false-action rates 0.056, 0.058 and 0.053, whereas the uncalibrated union serves 4,429, 4,231 and 4,180 at 0.125, 0.203 and 0.259: the union rule buys 0.40 versus 0.37 containment in $\mathcal{I}_{XFY}$ at more than four times the clean false-action rate and at a benign interruption of 0.76 versus 0.49. The earlier asymmetric rule served 4,494, 4,327 and 4,322; the correction moves the unsafe-allow count by less than 1% of the material rows. The unsafe allows are dominated by structure, not by thresholds: all 2,617 material label observations are served under $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ (residual blind region), 2,606 of them under batch-level $\mathcal{I}_{XFY}$ (statistically undetectable), and 1,057 of 1,079 material dropout observations under $\mathcal{I}_{XF}$; $\mathcal{I}_{Y_m}$ contains nothing in this suite.

The trusted-reference regime serves none of the 7,008 material observations with zero clean false actions on its 1,200 exact-zero rows (the aggregate reference alone would suffice for that, by Corollary 3a), and blocks every non-identity relabeling and every prediction change through its

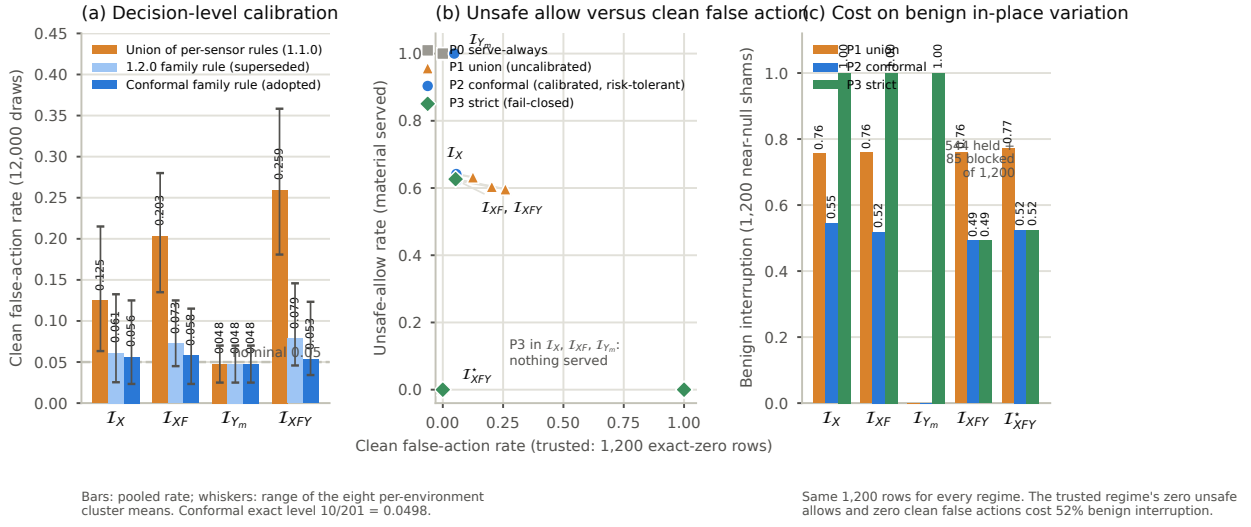


Fig. 1. Gates F and D (artifact 1.3.0). (a) Decision-level false-alarm rate on 12,000 disjoint clean draws of the union rule, of the superseded asymmetric rule and of the conformal family rule per regime; whiskers span the eight per-environment cluster means. (b) Unsafe-allow rate against clean false-action rate for every policy and regime at the primary materiality threshold; the trusted regime is the only point at the origin, its clean denominator is the 1,200 exact-zero rows, and the strict policy serves nothing where the label boundary is unverified. (c) Benign interruption: fraction of the same 1,200 near-null in-place controls held or blocked by each policy and regime; the trusted regime's origin in (b) costs 0.52 here.

TABLE 5

Offline end-to-end decisions on the frozen observations (primary materiality $|\Delta_R| > 0$; 7,008 material of 10,800 intervened observations). FPR: clean false-action rate on 12,000 disjoint clean draws, or on the 1,200 exact-zero rows for I_{XFY}^* (a different denominator). Benign: fraction of the 1,200 near-null in-place shams held or blocked (same rows for every regime). P0 serve-always allows all 7,008 material observations in every regime. Blind: material observations structurally indistinguishable from the reference under the regime.

Regime	Policy	FPR	Benign	Unsafe	Contain.	Blind
I_X	P1 union	0.125	0.76	4,429	0.37	2,617
I_X	P2 conf.	0.056	0.55	4,496	0.36	2,617
I_X	P3 strict	1.000	1.00	0	1.00	2,617
I_{XF}	P1 union	0.203	0.76	4,231	0.40	2,617
I_{XF}	P2 conf.	0.058	0.52	4,365	0.38	2,617
I_{XF}	P3 strict	1.000	1.00	0	1.00	2,617
I_{Y_m}	P1 union	0.048	0.00	7,008	0.00	5,450
I_{Y_m}	P2 conf.	0.048	0.00	7,008	0.00	5,450
I_{Y_m}	P3 strict	1.000	1.00	0	1.00	5,450
I_{XFY}	P1 union	0.259	0.76	4,180	0.40	0
I_{XFY}	P2 conf.	0.053	0.49	4,390	0.37	0
I_{XFY}^*	P2/P3 (exact)	0.000	0.52	0	1.00	0

item-aligned references. It is not free: of the 1,200 near-null in-place controls it holds 544 through its batch-level component and blocks 85 whose predictions changed under a 0.001 perturbation, a benign interruption of 0.52 against 0.52 for P2 in I_{XF} (Fig. 1(c)). Stronger provenance closes blind regions and incurs a specificity cost when exact invariance is enforced against benign variation; the two denominators of Table 5 are different and are reported side by side for that reason. The strict policy P3 makes information-set conditionality literal: in I_X , I_{XF} and I_{Y_m} the label boundary is unverified, so nothing is served and every clean and benign row is interrupted; in I_{XFY} and I_{XFY}^* it coincides with P2. The choice between P2 and P3 is therefore a declared risk decision, not a detector property. The composition with the six frozen contract envelopes reproduces their actions (supplement).

6.4 Adaptive Cluster-Preserving Attacker

Gate A ran as preregistered: 240 of 240 jobs, and the clean rows and matched controls reproduce the frozen expansion rows exactly. The attacker perturbs 0.32 of the entries on average (0.12–0.55 across environments, branches and dimensions). Fig. 2 and the supplement report the outcome. At the matched strengths 0.02, 0.05 and 0.10 the conformal rule detects the executed mean shift in 1.00, 1.00 and 1.00 of the rows in I_X and its cluster-preserving variant in 0.01, 0.08 and 0.34 (0.06, 0.24 and 0.48 in I_{XF}); scaling drift behaves alike (0.05–0.56 in I_X against 0.99–1.00). The attacker keeps most of the harm: the material fraction of the cluster-preserving mean shift is 0.29, 0.40 and 0.54 against 0.33, 0.46 and 0.59. Detection recovers at 0.25 and 0.50, where the unclustered entries move far enough to shift the feature distribution against the between-batch null. The classical branch (standardized features) is more evasive than the quantum branch (features scaled to $[0, 2\pi]$): at strength 0.10 the mean shift is detected in 0.30 and 0.34 of classical cells and 0.37 and 0.58 of quantum cells in I_X and I_{XF} ; the cluster rule is identical and the scaler sets how far the unclustered entries move. Of the 3,418 material adaptive rows, P2 serves 1,328 in I_X , 959 in I_{XF} and 1,006 in I_{XFY} , whereas the executed mechanisms in the same runs are served 0, 6 and 10 times out of 1,848; under I_{XFY}^* every material adaptive row is blocked, because a changed conclusion with fixed labels requires a changed prediction. The “detected in every cell” result for feature drift was a property of the fingerprint, not of the mechanism: calibration controls false alarms, it does not buy coverage against an attacker who knows what the sensor sees; only a reference does.

6.5 Quantum-Workflow Coverage

All 165 cells are present and all nine prespecified acceptance checks pass. Clean cells trigger no sensor. Benign transpilation and the common-unitary rewrite change serialized

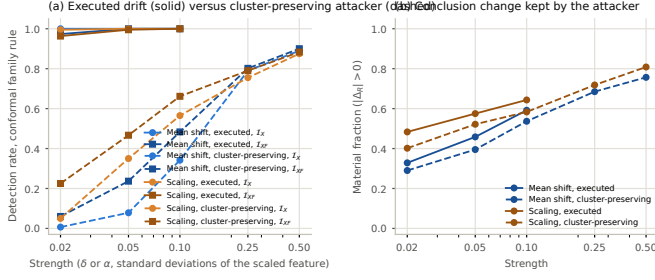


Fig. 2. Gate A. (a) Detection under the conformal family rule of the executed drift mechanisms (solid) and of their cluster-preserving variants (dashed) in $\mathcal{I}_X$ and $\mathcal{I}_{X_F}$, 600 observations per point. (b) Material fraction of the same rows: the attacker evades the batch-level sensors at matched strengths while keeping most of the conclusion changes.

provenance while preserving semantic kernels, algebraic properties and outputs in every cell (Proposition 7(i)): a raw hash difference needs policy-aware adjudication. Every data-dependent circuit mutation changes the semantic kernel; the mild RZ mutation changes no prediction, the stronger one changes predictions in 2/15 cells and the repetition change in 11/15. Asymmetric and diagonal-eroded matrices are detected algebraically in every cell and change no prediction; the PSD-preserving mixture is symmetric, unit-diagonal and PSD in all cells, so algebraic validation is exactly blind, while trusted provenance and semantic comparison expose all 15 substitutions and output monitoring reacts in 1/15 (Proposition 7(ii)). Both shot emulators produce non-zero repeated-estimation discrepancy in all cells and output changes in 7/15 and 3/15 (Fig. 3; Proposition 7(iii)). The prototype’s six envelopes verify and its eight acceptance checks pass: clean and approved transpilation are allowed, parameterized mutation and PSD-preserving substitution are blocked, prior-preserving label corruption is blocked at the evaluation contract despite a zero class-prior delta, and the approved 256-shot emulator is held. The secondary comparison between the ZZ kernel and the SVC, reported in full in the supplement, is retained only to show that the primary result does not depend on the choice of model: its direction is stable and its magnitude conditional on scalers and clean headroom, and it is neither a quantum-advantage nor a vulnerability ordering.

7 INTEGRITY-AUDIT CONTRACT

The evidence yields a reporting and enforcement contract: every integrity claim names the boundary and protected asset, the adversary or failure class and what remains fixed, the information regime and the trusted references it assumes, the sensor, the calibration rule and its decision-level level with the premise under which it holds, the covered interventions, the structural, calibrated and adaptive residual blind regions, the cost on benign variation, and the response. For the label path the minimum bundle is item-level label provenance plus a joint label-prediction audit; for the studied quantum boundary it is authenticated circuit/parameter provenance, an approved-transpilation policy, semantic probe kernels, algebraic checks, repeated-estimation evidence and abstention where no null is calibrated. The contracts fail closed on their invariants: a violated invariant blocks and missing

mandatory evidence holds. The decision policies differ by what they do with declared residual uncertainty: P2 may serve under an explicitly declared residual blind region; P3 interprets missing mandatory evidence as abstention. Neither converts missing evidence into a claim of integrity; P2 converts it into a declared, accepted risk.

8 DISCUSSION AND LIMITATIONS

8.1 Security Interpretation

A metric can move while predictor output is bit-for-bit unchanged, and it can move upward. Calling either model degradation assigns failure to the wrong component and suggests the wrong defence: input robustness cannot repair corrupted ground truth; provenance and joint validation can. Zero sensor response is informative only if the sensor receives evidence capable of distinguishing the states, and non-zero separability is informative only if the auditor holds a null tight enough to see it. The offline end-to-end result makes the consequence concrete: with the evidence a batch-level auditor normally has, about three fifths of the materially changed results in this suite would be served under a calibrated policy, and the share is set by the information regime, not by the threshold. The adversarial gate adds the second half of the lesson: a calibrated sensor family controls false alarms but not what an adaptive attacker can hide from it, so the “fully detected” mechanisms of the frozen suite are served in 28–39% of their material instances once the attacker preserves the fingerprint. The lever is provenance, and its granularity follows the integrity notion: a trusted aggregate of the same batch certifies the conclusion, item alignment certifies item identity, and nothing the adversary can rewrite certifies anything (Proposition 6). Provenance has a price that the evaluation measures rather than hides: exact invariance enforced against benign variation interrupts about half of the near-null controls.

8.2 Validity Boundaries

The study has the following limits. Three datasets and eight fixed scale/temporal settings, six of the nine environments from CICIDS2017, are not sampled deployments; five split clusters measure within-environment resampling only; most gates use 128/128 samples; balanced numeric-only staging does not reproduce operational prevalence. Both quantum evaluators are ideal statevector simulators with binomial shot emulation. SVC and QSVC keep pipeline-specific scalers, so their comparison does not isolate kernel geometry, and the same scalers govern how far an adaptive attacker’s unclustered entries move. Exact blindness is a capability result under constructed perturbations and estimates no prevalence. The conformal level is marginal and holds under exchangeability of the calibration draws and the audited batch; the executed design violates that premise (draws overlap within a pool half, the calibration and evaluation halves are different rows of one table, and E1’s pool half supplies one disjoint batch), and the residual excess is reported as executed, not hidden; no conditional guarantee given a calibration set is claimed. The adversarial gate fixes one cluster rule and two mechanisms; an attacker with a different model of the sensors is not evaluated. The trusted

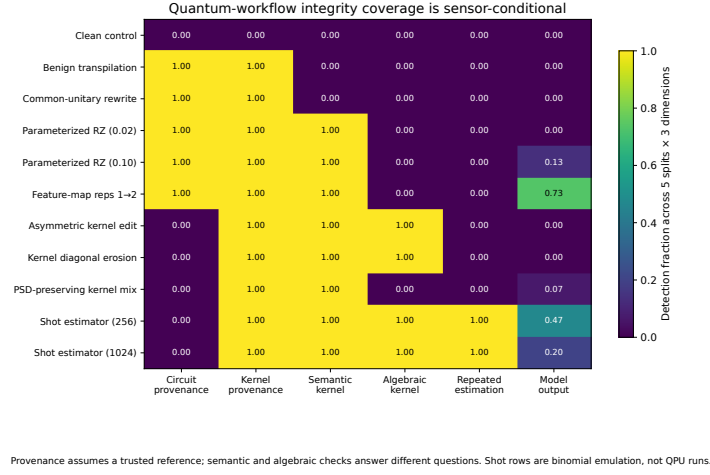


Fig. 3. Sensor response across clean and semantics-preserving controls, circuit mutations, kernel corruptions and finite-shot emulators over 165 simulator cells. Coverage is conditional on the evidence exposed to each sensor; output invariance does not imply workflow integrity.

item-aligned regime is a trust assumption, not a stronger detector: if the reference can be rewritten with the asset, Proposition 6 applies, and its benign-interruption cost is measured on near-null controls, not on operational variation. The prototype is a local process without authenticated provider records, and its hash chain certifies consistency of the record, not its authenticity. Every policy result is offline, on frozen outputs. Context-conditioned runtime calibration under non-stationarity, abstention with recovery, real-QPU execution and the service-level evaluation of enforcement are the subject of a separate study.

9 CONCLUSION

Hybrid workflow integrity cannot be reduced to a robustness number. An intervention can change a reported conclusion without changing the model, a sensor can stay silent because its view is identical, a separable change can stay undetected because the auditor holds no anchor, and a detected change can become undetected once the attacker learns what the sensor sees. We formalized this as observational indistinguishability under explicit information sets and trusted references, derived monotone and exactly closable blind regions with the label path as a protected boundary, placed the quantum branch on the same lattice, reproduced the boundaries across three datasets and eight environments, calibrated the decision with a conformal rule whose level is exact under a stated premise, and measured offline what each regime serves, what it costs on benign variation and how an adaptive attacker evades it. Every integrity claim must therefore state its boundary, view, trusted roots, decision-level level and premise, residual and adaptive blind regions, benign cost and response; assurance is granted only when the evidence the claim requires is present.

DATA, CODE, AND ETHICS STATEMENT

The review artifact contains source code, a locked environment, tests, compact derived tables, figures, seven SHA-256 evidence manifests, exact reproduction commands, the preregistrations of the 1.1.0–1.3.0 gates and the amendment

that replaced the calibration rule. Raw CICIDS2017, UNSW-NB15 and ToN-IoT data are not redistributed; their public sources, expected hashes and deterministic staging are documented. Version 1.3.0 is archived at Zenodo under the concept DOI 10.5281/zenodo.22550852 (version DOI in the artifact metadata); code is released under Apache-2.0, derived evidence, figures and documentation under CC BY 4.0. The study uses previously released benchmarks and involves no new human-participant or personal-data collection.

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